

**DYSARTHIC SPEECH RESYNTHESIS
USING DISCRETE DISENTANGLED SELF-
SUPERVISED REPRESENTATIONS**

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Dysarthria is a motor speech disorder that affects speech clarity due to muscle weakness or impaired motor control, resulting in slurred or unintelligible speech. While Automatic Speech Recognition (ASR) systems have been developed to assist dysarthric individuals, their effectiveness is limited by the scarcity of dysarthric speech data. To address this, the project proposes a voice conversion (VC)-based approach to enhance speech intelligibility and ASR performance. Initially, discrete disentanglement of speech features was explored to evaluate the feasibility of self-supervised speech units for resynthesis. Building on this, a Speech-to-Unit Translation (S2UT) model was developed to map dysarthric speech units to normal speech units, improving clarity while preserving speaker identity. A HiFi-GAN vocoder was used to synthesize high-quality speech waveforms. Subjective and objective evaluations confirmed significant improvements in intelligibility, while ASR experiments showed a substantial reduction in Word Error Rate (WER), validating voice conversion as an effective strategy for dysarthric speech enhancement and recognition.

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LIST OF ABBREVIATIONS

AAT	Augmentation Adversarial Training
ASR	Automatic Speech Recognition
CNN	Convolutional Neural Networks
CPC	Contrastive Predictive Coding
DL	Deep Learning
DNN	Deep Neural Networks
DMOS	Degraded Mean Opinion Score
EER	Equal Error Rate
F0	Fundamental Frequency
GAN	Generative Adversarial Network
k-NN	k- Nearest Neighbours
LM	Language Model
MFCC	Mel Frequency Cepstral Coefficient
ML	Machine Learning
MOS	Mean Opinion Score
SMA	Spinal Muscular Atrophy
S2UT	Speech to Unit model
TDSC	Tamil Dysarthric Speech Corpus
TDNN	Time Division Neural Network
TTDS	Text-to-dysarthric-speech
TTS	Text-to-speech system
VAE	Variational Auto Encoder
VQ	Vector Quantized
WER	Word Error Rate

CHAPTER 1

INTRODUCTION

Effective communication is the foundation of a fulfilling life, and speech plays a crucial role in expressing our thoughts and emotions. Dysarthria, a neurogenic speech disorder, significantly disrupts communication, which is a vital aspect of personal and social expression. Dysarthria comes in various types, each with its own set of perceptual attributes and underlying physiological causes. This chapter explores the primary types of dysarthria, delving into their unique auditory perceptual characteristics and the physiological mechanisms that contribute to their development. Additionally, it discusses existing treatment options for dysarthria, providing a foundation for the project's motivation and objectives.

1.1 DYSARTHRIA, A SPEECH DISORDER

Speech plays a crucial role in expressing the thoughts and feelings of an individual and plays a crucial role in individual participation and representation in society. Unfortunately, in India alone, more than 5 million people suffer from communication disabilities (Vikas Sheel, National Health Mission, New Delhi) and dysarthria affects 170 out of every 100,000 people in India (**Mahadik, 2019**).

Dysarthria is a neuromotor disorder affecting speech control, leading to decreased intelligibility. It results from muscular dysfunction in the five elements of speech: respiration, phonation, resonance, articulation, and prosody. Anarthria is a severe form with complete loss of speech production. While language comprehension is preserved, communication efficiency is compromised.

Dysarthria often involves drooling, feeding, and swallowing difficulties and significantly impacts individuals and their families due to its effects on communication, a vital aspect of personal and social expression. As a neurogenic speech disorder, dysarthria is characterized by abnormalities in the strength, speed, range, steadiness, tone, or accuracy of movements in the speech resonators and articulators that are required for breathing, phonatory, resonator, articulatory, or prosodic aspects of speech production **(Duffy, 2020)**.

A cornerstone of dysarthria research has been shaped fundamentally by identifying, discriminating, and characterizing different kinds of dysarthria. The differences in causation and consequences of dysarthria in different people have warranted this classification exercise. The primary types of dysarthria identified by the chief distinguishing auditory perceptual attributes and the associated localization of pathophysiology **(Duffy, 2020)** are highlighted in the following Table 1.1. The table can serve as a cogent overview of the kinds and characteristics of the types of dysarthria.

Table 1.1 Types, distinguishing features and pathophysiology of Dysarthria

DYSARTHRIA TYPE	Chief distinguishing auditory perceptual characteristics	Distinguishing pathophysiology
Flaccid	Breathiness; hoarseness; short phrases; audible inspiration; hypernasality; audible nasal emission; imprecise articulation	Weakness
Spastic	Strained-harsh voice quality; mono-pitch & mono loudness; slow rate	Spasticity

Ataxic	Irregular articulatory breakdowns; telescoping of syllables; distorted vowels; excess & equal stress; inappropriate variation in pitch, loudness & duration	Incoordination
Hypokinetic	Reduced loudness; mono-pitch & mono loudness; rapid or accelerated rate; short rushes of speech; reduced stress; repeated sounds; inappropriate silences	Rigidity; bradykinesia; reduced range of motion; movement scaling problems
Hyperkinetic	Variable, dependent on the locus in speech mechanism of fast-to-slow and rhythmic-to-unpredictable movements	Involuntary movements (e.g., chorea, myoclonus, tics, dystonia, tremor)

Table 1.1 adapted from **(Duffy, 2014)** expounds on the main types of dysarthria and their manifestations as specific characteristics of disordered speech. An individual with dysarthria mainly exhibits weakness of muscles in the facial region, restricted movement of the tongue, the lips and the jaw, difficulty breathing, change in voice quality (hoarse, breathy or strained voice), monotone “robotic” speech or extreme change in pitch, mumbling or slurred speech, slow rate of speech, nasal sounding voice, 4 imprecise pronunciation of speech sounds, rapid deterioration of speech quality, intelligibility and prosody and difficulty chewing and swallowing.

1.2 VOICE CONVERSION

Voice conversion (VC) is a speech-processing technique that modifies the voice characteristics of a source speaker to match those of a target speaker while preserving linguistic content **(Childers & Wu, 1990)**. It is widely used in applications such as speech synthesis, dubbing, anonymization, and speaker adaptation **(Kain et al., 1998; Stylianou,**

2009). Traditional VC methods relied on statistical models like Gaussian Mixture Models (GMMs) and Hidden Markov Models (HMMs), which transformed spectral features but often resulted in unnatural speech (**Toda et al., 2012**). With advancements in deep learning, modern voice conversion techniques leverage deep neural networks (DNNs), variational autoencoders (VAEs), generative adversarial networks (GANs), and self-supervised models such as Wav2Vec and HuBERT (**Polyak et al., 2021**). These approaches extract speaker-independent content representations while disentangling speaker identity, pitch, and prosody, enabling high-quality voice transformation. More recent unit-based methods eliminate the need for text labels, making voice conversion feasible even in low-resource languages (**Lee et al., 2022**).

In the context of dysarthric speech, voice conversion can be adapted as a powerful tool to improve intelligibility by modifying the acoustic properties of speech while preserving the speaker's identity. Voice conversion addresses this issue by mapping dysarthric speech features onto a healthier acoustic space, effectively enhancing clarity, phoneme distinction, and naturalness without altering the linguistic content (**Wang et al., 2022**). A key advantage of voice conversion in dysarthric speech is its ability to retain speaker identity while improving speech quality. This is particularly important for individuals who rely on assistive communication technologies, as preserving their unique vocal characteristics ensures familiarity and personal expression (**Veaux et al., 2016**). Unlike traditional speech enhancement methods that focus solely on noise reduction or spectral modifications, voice conversion learns to correct distortions at multiple levels, including articulation, prosody, and phonation (**Toda et al., 2012**). Advanced neural architectures such as sequence-to-sequence (seq2seq) Transformers and deep generative

models can effectively separate speaker-independent content from dysarthric speech while reconstructing a more intelligible version that sounds natural and fluent (**Wang et al., 2020**).

1.3 DATA AUGMENTATION FOR DYSARTHRIA

Owing to the debilitating nature of this condition, many data augmentation techniques have been developed. **Matsuzaka et al., (2015)** proposed a deep neural network (DNN)-based Text-to-Speech (TTS) model to synthesize dysarthric speech. The process involved training a TTS model on a small dataset of recorded dysarthric speech and then generating additional synthetic speech for ASR model training. This method addresses the challenge of limited data by augmenting the dysarthric speech data, ultimately enhancing the model’s recognition accuracy. Through experiments on spinal muscular atrophy (SMA) patients, the study demonstrates that augmenting the training dataset with synthesized dysarthric speech improves speech recognition performance, providing a viable solution for augmenting dysarthric data without burdening patients with extensive recordings.

Conversely, **Leung et al., (2019)** introduced a diffusion probabilistic model (DPM) for text-to-dysarthric-speech (TTDS) synthesis, specifically utilizing Grad-TTS. This model synthesizes dysarthric mel-spectrograms to create augmented datasets for ASR fine-tuning, especially focusing on ASR adaptation for severe dysarthria. By employing large-scale ASR foundation models such as Whisper, this study achieves a more nuanced representation of dysarthric characteristics, effectively capturing speaker-specific features through a diffusion process. The results indicate that DPM-based TTDS can significantly enhance ASR performance metrics, including Word Error Rate (WER),

across a spectrum of dysarthric severity levels, thus offering a promising direction for severe dysarthria data augmentation.

Transformer-based models have not been extensively explored for dysarthric speech, as such architectures typically require substantial training data that is often unavailable publicly. **(Almadhor et al., 2023)** Implemented a spatial convolutional neural network (CNN) with multi-head Transformers, pre-trained on control speaker data, to recognize visual representations of whole-word dysarthric speech. **(Shahamiri et al., 2021)** Used dysarthric and typical speech corpora along with data augmentation techniques to perform two-step parameter adjustments for training a dysarthric Transformer model. Additionally, a recurrent neural network (RNN)-Transducer model was trained on the Euphonia dataset. Recently, representations from foundational ASR models, such as Wav2Vec2.0 **(Hernandez et al., 2022)** and WavLM **(Violeta et al., 2022)**, have been used as input features for dysarthric ASR systems, although these foundation models have yet to be adapted specifically for dysarthric ASR.

1.4 MOTIVATION AND OBJECTIVES

The motivation behind voice conversion for dysarthric speech stems from the critical need to improve communication for individuals with speech impairments. Existing speech recognition systems and assistive communication technologies struggle to process dysarthric speech effectively, primarily due to the high variability and limited availability of dysarthric speech data. Traditional speech enhancement techniques often fail to generalize across different severities of dysarthria, resulting in inconsistent improvements in intelligibility. Therefore, there is a pressing need for a robust, adaptive method that can enhance speech

clarity while preserving speaker identity, ensuring that individuals with dysarthria can communicate more effectively without losing their unique vocal characteristics.

The objective of this work is to develop a **textless dysarthric voice conversion system** that transforms disordered speech into a clearer, more natural form while maintaining speaker individuality. Unlike conventional voice conversion approaches that rely on text or phonetic transcriptions, this system utilizes soft-unit-based speech representations to preserve fine articulatory details without requiring linguistic supervision. The proposed framework employs a seq2seq Transformer-based Speech-to-Unit Translation (S2UT) model to convert dysarthric speech into an intelligible representation, followed by a HiFi-GAN vocoder to synthesize high-quality speech. Additionally, the system serves as an effective data augmentation strategy for dysarthric Automatic Speech Recognition (ASR) by generating intelligible training data, reducing the Word Error Rate (WER) significantly. By leveraging self-supervised learning and soft-unit modeling, this approach provides a scalable, low-resource-solution to improve dysarthric speech processing, enabling more effective human-machine and human-human communication for individuals with speech impairments.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

This literature review examines key advancements in dysarthric speech recognition and enhancement, with a particular focus on voice conversion techniques for improving intelligibility. The review begins with **Kain et al. (2007)** and **Toda et al. (2012)**, who introduced early methods for improving speech intelligibility. More recent studies by **Jiao et al. (2018)** and **Vachhani et al. (2018)** focused on data augmentation techniques to enhance ASR robustness for dysarthric speech. Further advancements in intelligibility enhancement were explored by **Wang et al. (2020)** and **Polyak et al. (2021)**, leveraging deep learning-based resynthesis approaches. Subsequent works by **Wang et al. (2024)** and **van Niekerk et al. (2022)** introduced unit-based voice conversion techniques, addressing phonetic degradation challenges. Inspired by textless speech-to-speech translation, **Lee et al. (2022)** applied unit-based speech transformations for improved generalization. Additionally, **Celin et al. (2023)** investigated transfer learning strategies to improve dysarthric ASR performance. These studies collectively contribute to the evolving landscape of dysarthric speech processing, showcasing diverse methodologies in intelligibility enhancement, data augmentation, and voice conversion.

2.2 COMPREHENSIVE LITERATURE REVIEW

Kain et al. (2007) introduced one of the earliest approaches to dysarthric speech enhancement using spectral voice conversion

techniques. Their method involved transforming the spectral features of dysarthric speech to better match those of a target healthy speaker, thereby improving intelligibility. The proposed system utilized Gaussian Mixture Models (GMMs) to learn the mapping between dysarthric and normal speech spectral parameters, enhancing articulation clarity while maintaining speaker identity. Although this approach demonstrated improvements in speech quality, it was limited by its reliance on parallel speech data, making it difficult to apply in real-world scenarios where such data is scarce.

Toda et al. (2012) further advanced the statistical approach by proposing a Maximum Likelihood Estimation (MLE)-based spectral parameter trajectory method for voice conversion. This method improved upon earlier GMM-based models by ensuring smoother spectral transitions, thereby producing more natural-sounding speech. The study highlighted the effectiveness of MLE in refining phonetic details in dysarthric speech, which is crucial for maintaining intelligibility. However, the method still faced challenges related to over-smoothing effects and the loss of fine-grained articulatory nuances, limiting its real-world applicability.

Jiao et al. (2018) explored Generative Adversarial Networks (GANs) for generating dysarthric-like speech from healthy speakers to augment training data for ASR systems. The study aimed to address the data scarcity problem in dysarthric ASR by leveraging GANs to synthesize speech that mimicked the acoustic characteristics of dysarthria. This approach significantly improved ASR robustness for dysarthric speakers by expanding the training dataset with synthetic but realistic dysarthric utterances. Despite these improvements, GAN-generated speech

sometimes introduced unnatural artifacts, which could affect model generalization.

Vachhani et al. (2018) proposed a spectral and prosodic perturbation-based data augmentation technique for dysarthric speech recognition. Unlike GAN-based methods, their approach involved modifying formants, pitch, and energy distributions to create diverse dysarthric-like variations from a limited dataset. The study showed that adding these perturbed speech samples to the ASR training pipeline significantly reduced Word Error Rate (WER). However, while this method improved ASR generalization, it did not directly enhance the intelligibility of real dysarthric speech.

Wang et al. (2020) introduced an end-to-end voice conversion model that utilized cross-modal knowledge distillation to improve dysarthric speech intelligibility. The system leveraged a seq2seq speech transformation model to convert dysarthric utterances into clearer speech representations, enhancing phonetic structure while preserving speaker identity. The key innovation of this study was the integration of linguistic and acoustic priors from a teacher ASR model, allowing for more effective speech reconstruction. The approach showed significant reductions in WER for dysarthric ASR, demonstrating the benefits of resynthesized speech for both intelligibility and recognition.

Polyak et al. (2021) proposed a disentangled representation learning framework for speech resynthesis, where speech was broken down into three distinct components: content, pitch (F0), and speaker identity. Their method leveraged self-supervised learning (SSL) to extract these features separately, enabling precise control over speech

modification. This approach proved effective in improving voice quality and phonetic accuracy while preserving essential speaker characteristics. However, the reliance on discrete speech units introduced quantization artifacts, which sometimes degraded naturalness.

Wang et al. (2024) introduced Unit-DSR, a dysarthric speech reconstruction system that employed HuBERT-based discrete unit modelling followed by a unit-based HiFi-GAN vocoder. The approach aimed to transform dysarthric speech into an intelligible form while maintaining linguistic accuracy. Unit-DSR demonstrated notable improvements in phoneme articulation and prosody correction, significantly reducing WER in dysarthric ASR tasks. However, since the method relied on discrete speech units, it suffered from quantization errors, which impacted fine phonetic details.

van Niekirk et al. (2022) conducted a study comparing hard (discrete) and soft (continuous) speech units for voice conversion. Their findings revealed that while discrete units (such as those used in HuBERT and VQ-VAE models) provided better speaker normalization, they often degraded phonetic clarity due to quantization errors. In contrast, soft units, which are continuous probabilistic representations, preserved articulatory nuances more effectively, leading to higher intelligibility in voice conversion tasks. This study laid the foundation for adopting soft-unit-based speech processing, particularly for dysarthric voice conversion.

Lee et al. (2022) explored textless speech-to-speech translation models, demonstrating that unit-based representations could enable effective speech conversion without requiring text transcriptions. Their approach was particularly relevant for low-resource languages, where

large-scale parallel speech datasets are unavailable. By mapping speech directly from one unit representation to another, the study proved that non-parallel, text-independent voice conversion was feasible, opening new possibilities for dysarthric speech enhancement.

Celin et al. (2023) investigated transfer learning techniques to improve ASR performance for dysarthric speech. Their study incorporated synthetically generated dysarthric speech into the ASR training pipeline, significantly boosting recognition accuracy. The findings reinforced the idea that resynthesized dysarthric speech could serve as valuable training data, reducing ASR error rates and improving model robustness across different severities of dysarthria.

2.3 SUMMARY

In summary, this literature review presents a comprehensive overview of the advancements in dysarthric speech recognition and enhancement methodologies. Early approaches, such as those by **Kain et al. (2007)** and **Toda et al. (2012)**, established the foundation for dysarthric speech modification through statistical parametric methods, improving spectral clarity while preserving speaker identity. **Jiao et al. (2018)** and **Vachhani et al. (2018)** explored data augmentation techniques, including GAN-based augmentation and spectral perturbation, to enhance ASR robustness for dysarthric speech.

Further advancements in intelligibility enhancement were introduced by **Wang et al. (2020)**, who applied cross-modal knowledge distillation for dysarthric speech resynthesis, and **Polyak et al. (2021)**, who developed a speech resynthesis framework using discrete disentangled representations.

More recently, **Wang et al. (2024)** proposed Unit-DSR, leveraging HuBERT-based discrete unit modelling to improve dysarthric speech intelligibility, while **van Niekerk et al. (2022)** compared hard and soft speech units, highlighting the advantages of continuous probabilistic representations for preserving articulatory nuances. Inspired by **Lee et al. (2022)**, textless speech-to-speech translation methods have demonstrated the effectiveness of unit-based conversion for low-resource languages. Additionally, **Celin et al. (2023)** investigated transfer learning techniques, reinforcing the importance of synthetic dysarthric speech for ASR performance improvement.

Building on these methods, the current work develops a textless dysarthric voice conversion system that integrates soft-unit-based speech representations with a HiFi-GAN vocoder to synthesize intelligible dysarthric speech while maintaining speaker characteristics. This approach employs seq2seq speech-to-unit translation (S2UT) to enhance phonetic clarity without relying on transcriptions. Additionally, the resynthesized speech serves as augmented training data for dysarthric ASR, significantly reducing Word Error Rate (WER).

CHAPTER 3

SPEECH RESYNTHESIS FOR DYSARTHIC SPEECH ENHANCEMENT

3.1 INTRODUCTION

Disentanglement in speech processing involves decomposing complex speech signals into distinct components such as linguistic content, speaker identity, and pitch. This technique plays a crucial role in speech synthesis and transformation tasks, as it enables better control over individual speech attributes while preserving naturalness. Recent advancements in self-supervised learning have significantly improved the ability to extract and manipulate these speech components from large-scale, unlabelled datasets. By leveraging learned representations, speech disentanglement techniques have facilitated improvements in voice conversion, speaker adaptation, and intelligibility enhancement, particularly benefiting individuals with speech impairments.

This project explores two distinct methodologies for dysarthric speech resynthesis, both focusing on disentangling speech into interpretable units but differing in their approach. The first methodology, discrete disentanglement of speech features, serves as a proof of concept, demonstrating that HuBERT-derived discrete units can be effectively used to reconstruct intelligible speech. This method validates the potential of discrete representations in capturing essential speech features while allowing for high-quality resynthesis. Building on this foundation, the second methodology, voice conversion based on discrete units, refines the approach by integrating soft-unit modelling and a seq2seq Speech-to-Unit Translation (S2UT) framework. This enhanced method not only improves

speech intelligibility but also ensures better preservation of speaker identity and prosody, leading to a more natural resynthesis of dysarthric speech.

This chapter will explain both methodologies, highlighting their differences and improvements.

3.2 DATASET DESCRIPTION

The Tamil Dysarthric Speech Corpus (TDSC), also known as SSNCE-TDSC, is a specialized speech dataset designed to facilitate research in dysarthric speech processing. Developed by the Speech Lab, SSN College of Engineering (Mariya Celin et al., 2016), the corpus consists of speech recordings from 30 speakers, including 20 dysarthric individuals (13 male, 7 female) and 10 healthy speakers (5 male, 5 female). The dysarthric speakers, diagnosed with cerebral palsy (spastic quadriplegia or bilateral paraplegia), range in age from 12 to 37 years. Each speaker contributed 365 utterances, covering 103 isolated words and 262 sentences containing both common and uncommon Tamil phrases. This structure ensures comprehensive phonetic coverage of the Tamil language.

The recordings were collected in a controlled laboratory environment in collaboration with the National Institute for the Empowerment of Persons with Multiple Disabilities (NIEPMD). A head-mounted microphone was used to capture high-quality audio at a sampling rate of 16 kHz, ensuring consistency across all recordings. In total, the corpus comprises approximately eight hours of speech data, accompanied by time-aligned transcripts and metadata.

For this study, we focused on a subset of the dataset, selecting only mild and moderate dysarthric speakers, totalling 17 dysarthric participants, while excluding severe cases to ensure more stable unit alignments. The healthy speaker subset, comprising 3,650 utterances (10 speakers $\times$ 365 utterances). Dysarthric speech served as input to the Speech to-Unit Translation (S2UT) model, with target outputs derived from the corresponding healthy speaker’s soft units. The first methodology explores the feasibility of using HuBERT-derived discrete units for speech reconstruction, serving as a proof of concept for unit-based speech resynthesis. As shown in Figure 1, the pipeline begins with extracting discrete speech units from the original speech using a HuBERT-based Speech-to-Unit model, pre-trained on normal speech. Simultaneously, a speaker embedder (ECAPA-TDNN) extracts speaker identity features from the input speech. The discrete units are then passed through a pre-initialized soft encoder, which converts them into soft continuous representations before being fed into an acoustic model for refinement. Finally, the HiFi-GAN vocoder, equipped with convolutional dilated residual blocks, synthesizes the final resynthesized speech output. This methodology validates that HuBERT units retain essential speech information and can be effectively leveraged for speech reconstruction, forming the basis for further improvements in voice conversion.

3.3 SPEECH RESYNTHESIS FROM DISCRETE REPRESENTATIONS

The first methodology explores the feasibility of using HuBERT-derived discrete units for speech reconstruction, serving as a proof of concept for unit-based speech resynthesis. As shown in Figure 3.1, the pipeline begins with extracting discrete speech units from the original

speech using a HuBERT-based Speech-to-Unit model, pre-trained on normal speech. Simultaneously, a speaker embedder (ECAPA-TDNN) extracts speaker identity features from the input speech. The discrete units are then passed through a pre-initialized soft encoder, which converts them into soft continuous representations before being fed into an acoustic model for refinement. Finally, the HiFi-GAN vocoder, equipped with convolutional dilated residual blocks, synthesizes the final resynthesized speech output. This methodology validates that HuBERT units retain essential speech information and can be effectively leveraged for speech reconstruction, forming the basis for further improvements in voice conversion.

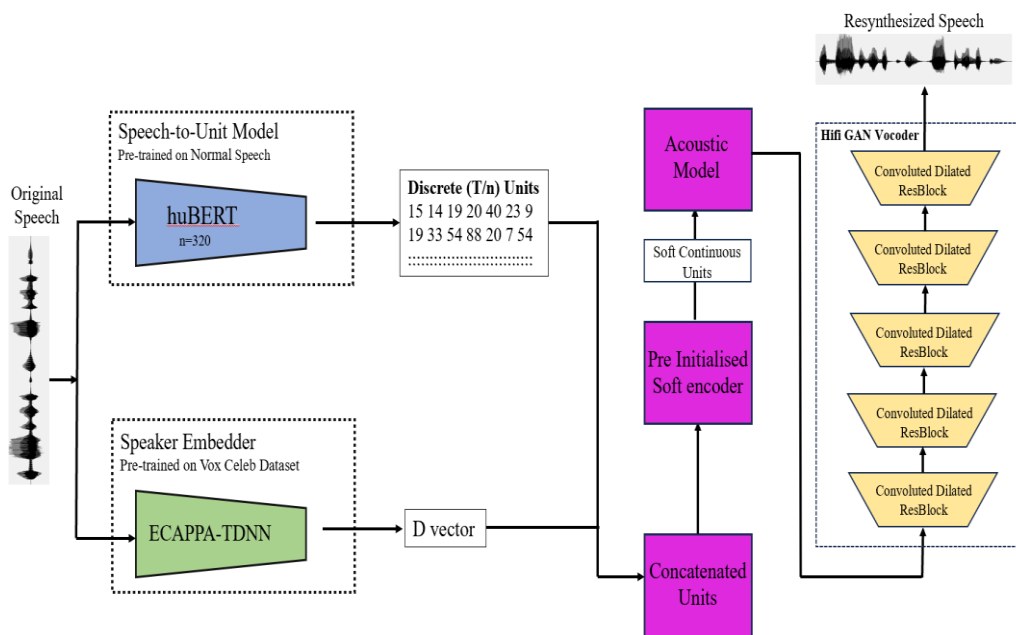


Figure 3.1 Proposed Discrete Disentangled Dysarthric speech resynthesis System

3.3.1 Speech – Unit Model

This section explains the key components and processes involved in the Speech-to-Unit Model, which is essential for extracting meaningful representations from raw speech data. The model transforms speech signals into discrete units, which can then be used for downstream tasks like voice conversion and automatic speech recognition (ASR). In the subsequent subsections, we explore the architecture of HuBERT, a self-supervised model used for speech unit extraction, and the process of extracting these units using HuBERT.

3.3.1.1 HuBERT Architecture

HuBERT (Ning Hsu. et al, 2021) as shown in Fig 3.2 is a self-supervised model designed for speech representation learning by predicting hidden units from masked segments of audio input. The architecture includes a CNN encoder, Transformer layers, and an acoustic unit discovery system that leverages unsupervised clustering (e.g., K-means on MFCCs) to generate pseudo-labels for speech units. Initially, the CNN encoder extracts low-level features from the raw audio waveform, which are then masked in specific spans to prevent overfitting to local details. The Transformer network processes these masked inputs to learn contextual representations, with the model trained to predict the cluster labels of the masked segments. This forces HuBERT to capture both the acoustic features and the broader linguistic structure of speech.

The HuBERT model's iterative training includes multiple rounds of clustering and refinement to enhance representation quality. The masked prediction objective, inspired by BERT, focuses on predicting only masked regions, ensuring that the unmasked portions help the model learn

contextual dependencies. Through this approach, HuBERT can achieve a comprehensive understanding of speech without requiring labelled data, enabling it to excel in low-resource scenarios. This makes HuBERT a versatile and robust tool for various downstream speech tasks.

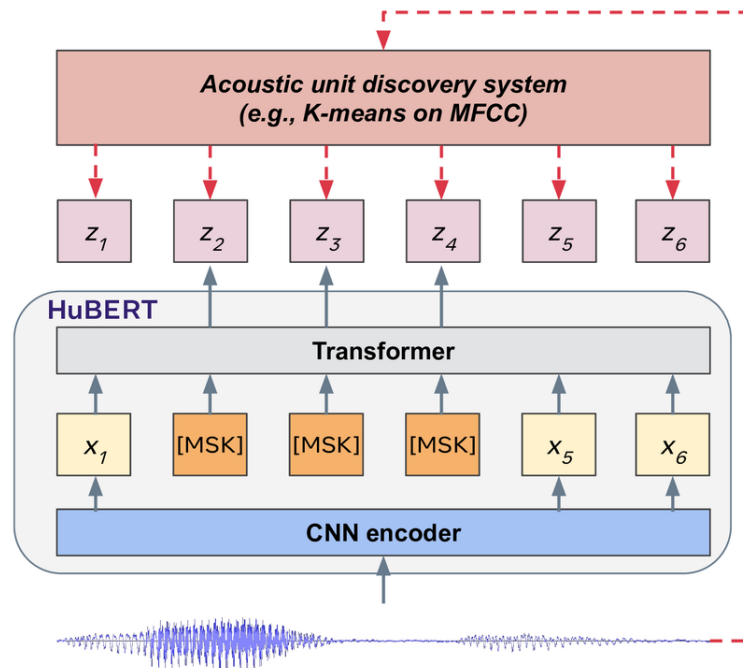


Figure 3.2: HuBERT Architecture

3.3.1.2 Extraction of speech units from HuBERT

In this project, we have used Hubert Model to extract the speech units. Figure 3.3 illustrates the process of transforming raw audio data into discrete speech units, which are then used as "pseudo-labels" in self-supervised learning for models like HuBERT. The process begins with the feature extractor, which converts the raw audio waveform into acoustic features such as Mel-frequency cepstral coefficients (MFCCs), Contrastive Predictive Coding (CPC), or representations derived from models like HuBERT. These features capture essential characteristics of the audio that are critical for identifying speech sounds. By using

representations that reflect the structure of the sound spectrum, the feature extractor produces a more structured and meaningful representation of the audio input, making it easier for the model to learn distinctions between different speech sounds.

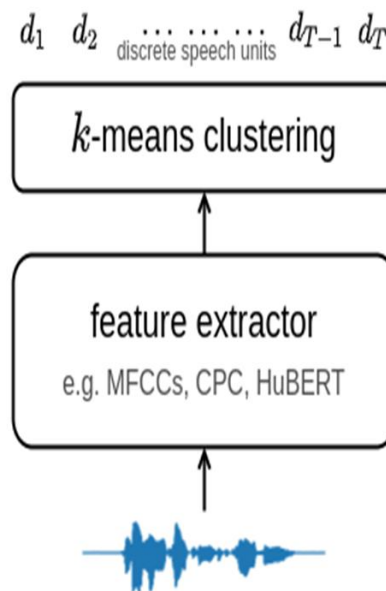


Figure3.3: Units Extraction using HuBERT

Following feature extraction, k-means clustering is applied to the extracted features to group similar frames into discrete speech units, labeled here as $(d_1, d_2, \dots, d_T)$. Each cluster represents a unique unit of sound, capturing specific phonetic or acoustic characteristics that are inherent in the speech data. These clusters act as "pseudo-labels". The discrete units obtained from clustering are used as target labels in the training process, allowing the model to learn meaningful audio representations by predicting the correct unit for each segment. This approach enables the model to focus on capturing both local acoustic details and broader contextual patterns, which are essential for robust speech representation in low-resource or unannotated data settings.

3.3.2 Speaker Encoder

In this section, we delve into the Speaker Encoder, a critical component in the voice conversion pipeline. The Speaker Encoder plays a vital role in capturing speaker-specific characteristics from the input speech, which are essential for maintaining speaker identity during the speech enhancement process. The architecture used for this task is the ECAPA-TDNN, which improves feature extraction through attention layers and attention mechanisms. The following section explores the architecture of ECAPA-TDNN and its contributions to speaker verification tasks.

3.3.2.1 ECAPA Architecture

The ECAPA-TDNN architecture as shown in Fig. 3.4 is an advanced variant of the Time Delay Neural Network (designed for speaker verification tasks). It addresses some limitations of traditional TDNNs by introducing sophisticated modules that enhance feature extraction and embedding quality. One of the key advantages in ECAPA-TDNN is the Channel- and Context-Dependent Statistics Pooling layer. This layer replaces standard pooling with a channel-specific attention mechanism that computes statistics across both local and global contexts. By allowing each channel to focus on different parts of the time sequence based on speaker-specific information, the model can better capture variations in speech. This focus on relevant speech segments, rather than a uniform pooling approach, provides a richer and more representative embedding that is useful in distinguishing speakers.

Another core enhancement in the ECAPA-TDNN architecture is the integration of 1-Dimensional Squeeze-Excitation Res2Blocks. These

blocks combine elements from the Res2Net module with 1D SE blocks to model multi-scale features across temporal dimensions effectively. The SE mechanism applies global channel attention, where each channel is weighted based on the importance of its features across the entire utterance. By dynamically adjusting the attention given to each channel, the SE blocks emphasize more informative features while suppressing less relevant ones. This attention improves the model's ability to extract nuanced speaker characteristics from the audio, and the residual connections within the Res2Blocks allow the network to capture a broad temporal context without overfitting or increasing the parameter count.

Additionally, ECAPA-TDNN employs a Multi-layer Feature Aggregation and Summation strategy that leverages outputs from multiple layers rather than relying solely on the final layer. By aggregating information across different layers, the architecture combines both shallow, fine-grained details with deeper, more abstract features. This hierarchical aggregation is achieved by summing outputs from each SE-Res2Block, ensuring that important speaker information from earlier layers is preserved and integrated with high-level representations.

This approach provides a more comprehensive embedding that is resilient to variations in speaker characteristics, leading to better generalization. Together, these architectural improvements enable ECAPA-TDNN to achieve state-of-the-art results on speaker verification benchmarks, offering a powerful model for applications that require reliable speaker identification.

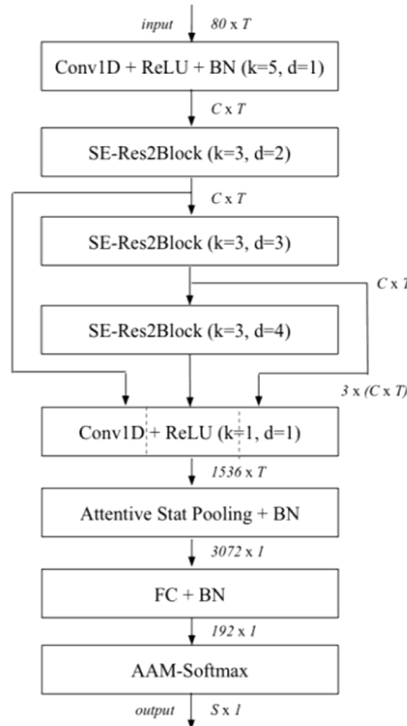


Figure 3.4 ECAPA-TDNN Network

3.3.2.2 Extraction of d- vectors

In the ECAPA-TDNN network, d-vectors are extracted as a way to represent speaker-specific characteristics from audio input. The process begins with the input audio signal passing through several initial layers, including convolutional and Time Delay Neural Network (TDNN) layers, to capture temporal patterns in the speech. These layers are designed to model short- and long-term dependencies, enabling the network to understand the structure of speech over time.

Next, these features are refined through the 1D Squeeze-Excitation (SE) Res2Blocks within the ECAPA-TDNN architecture. The SE-Res2Blocks enhance the feature maps by applying a channel-wise attention mechanism, dynamically weighting different channels based on

the global context of the utterance. This mechanism enables the network to focus on channels that carry the most speaker-discriminative information. Additionally, the Res2Net-inspired structure allows multi-scale feature extraction, so the network can capture details at different granularities, further enriching the feature representation. The result of this stage is a set of features that are not only temporally aware but also emphasize speaker-specific information, which is critical for accurate d-vector extraction.

Finally, the Channel- and Context-Dependent Statistics Pooling layer aggregates the refined features across the entire input sequence, creating a single, fixed-size embedding that represents the speaker. This layer computes statistics like the mean and standard deviation for each channel, emphasizing key aspects of the speaker's voice characteristics across time. By using context-aware pooling, this step ensures that the final embedding captures comprehensive speaker information while reducing noise and irrelevant details. The aggregated feature is then passed through a final fully connected layer, resulting in the d-vector. This d-vector is a robust, fixed-dimensional representation of the speaker's voice, making it suitable for tasks like speaker verification, where unique and stable speaker embeddings are essential for accurate identification.

3.3.3 Soft Unit Encoder

After concatenating the discrete speech units (content, pitch, and speaker identity) with the D-vector, the resulting representation is processed through a pre-initialized soft encoder. The purpose of this transformation is to convert discrete units into soft continuous representations, which better capture the articulatory details and phonetic nuances of speech. Unlike purely discrete units, which can suffer from

quantization errors, soft units provide a probabilistic mapping that helps retain the subtle variations in pronunciation and speech dynamics.

The soft content encoder is trained to predict a probability distribution over discrete speech units for each time step. It utilizes a self-supervised backbone model, such as HuBERT or Contrastive Predictive Coding (CPC), to extract robust speech features. A linear projection layer is then applied to map these extracted features into soft-unit representations. Each soft unit is represented as a probability distribution as shown in Eqn.1, which is computed using cosine similarity between the discrete and soft units, scaled by a temperature parameter (τ). The probability of a discrete unit $d_t = i$ given a soft unit s_t is defined as:

$$p(d_t = i | s_t) = \frac{\exp(\sin(s_t, e_i) | \tau)}{\sum_{k=1}^K \exp(\sin(s_t, e_k) | \tau)} \quad (1)$$

where e_i represents the discrete unit embeddings and K is the total number of units. This ensures that the soft units are smooth interpolations of discrete speech features rather than rigid categorical assignments.

The soft encoder is trained using cross-entropy loss, which minimizes the difference between the predicted soft-unit distributions and the corresponding discrete unit targets. This allows the encoder to generalize across different speakers and speech variations, ensuring that the learned representations are robust and capture phonetic and speaker-specific information effectively.

3.3.4 Acoustic Model

The acoustic model plays a crucial role in transforming soft continuous speech units into meaningful representations that guide the final waveform synthesis. It acts as an intermediary between the pre-initialized soft encoder and the HiFi-GAN vocoder, ensuring that the phonetic, prosodic, and speaker-specific attributes are effectively modeled before audio reconstruction. The input to the acoustic model consists of soft units derived from discrete representations, which provide a smooth probabilistic mapping of speech content, pitch, and speaker identity. These units are refined by the acoustic model to ensure that the generated representations capture long-term speech dependencies, improving the naturalness and intelligibility of the synthesized speech.

The acoustic model operates as a seq2seq framework that maps the input soft speech units to a refined latent space, making them more interpretable for the HiFi-GAN vocoder. It employs convolutional and recurrent layers to process speech sequences efficiently, capturing both short-term phonetic details and long-term speech rhythm patterns. Unlike traditional TTS acoustic models that rely on explicit text supervision, this model learns speech transformations directly from unit-based representations, making it particularly effective for dysarthric speech resynthesis, where textual alignment is challenging. Furthermore, the model is optimized using cross-entropy loss, which ensures that the predicted unit representations remain close to their ideal targets, enhancing pronunciation clarity and speaker preservation. Once processed by the acoustic model, the refined speech units are passed to the HiFi-GAN vocoder, which decodes these representations into high-fidelity waveforms.

3.3.5 Decoder

Figure 3.1 shows the modified HiFi-GAN neural vocoder used in this study is used to decode discrete speech representations into high-quality audio waveforms. It is composed of a generator, denoted as G , and a set of discriminators, denoted as D , which work in tandem to synthesize natural-sounding speech. The generator receives the encoded representation of the speech, which includes discrete representations of content, pitch, and speaker identity. The generator's role is to upsample these discrete representations to the target audio sample rate while ensuring that the synthesized audio captures the nuances of the original speech characteristics. By combining multiple architectures, the HiFi-GAN vocoder can resynthesize the speech.

In the generator, the discrete content and pitch representations are first passed through a set of look-up tables (LUTs) that transform these discrete inputs into continuous embeddings. These embeddings are crucial for smooth and continuous speech synthesis, as they allow the generator to produce a natural-sounding waveform. The generator includes a series of transposed convolution layers that progressively upsample the encoded features, increasing their temporal resolution to match the target sampling rate. Additionally, each transposed convolution layer is followed by a residual block with dilated layers, which expand the receptive field without additional computational burden. This expanded receptive field enables the generator to capture long-term dependencies, making the synthesized speech coherent over extended durations.

The encoded inputs to the generator consist of three key components: the discrete content sequence z_c , the pitch sequence z_{F0} , and

the speaker embedding z_{spk} . The content and pitch sequences are first transformed into continuous vectors by the respective look-up tables (LUT_c and LUT_{F0}), upsampled, and concatenated to form a unified representation. The speaker embedding z_{spk} is then concatenated to each frame of the upsampled sequence, allowing the generator to incorporate speaker-specific characteristics into the output. This concatenation process ensures that the synthesized speech reflects the unique vocal traits of the target speaker, resulting in a more personalized and realistic audio output.

To ensure the quality of the synthesized audio, the HiFi-GAN vocoder employs two types of discriminators: the Multi-Period Discriminator (MPD) and the Multi-Scale Discriminator (MSD). The MPD is specifically designed to analyze the generated audio by evaluating patterns over specific periodic intervals, helping to capture rhythmic qualities in the speech. It consists of multiple sub-discriminators, each operating at different periodic intervals to examine different rhythmic structures within the audio. By using intervals of [2, 3, 5, 7, 11], the MPD can detect a variety of periodic patterns, which is essential for capturing the natural rhythm and flow of speech.

The Multi-Scale Discriminator (MSD) complements the MPD by focusing on the frequency structure of the generated audio at various scales. The MSD consists of sub-discriminators that operate at different downsampled versions of the audio: the original scale, a downsampled scale by a factor of 2, and a downsampled scale by a factor of 4. Each sub-discriminator in the MSD captures details at a different level of granularity, enabling the model to capture both fine-grained and broad patterns within the audio. By analyzing the audio at multiple scales, the

MSD effectively captures short- and long-term dependencies, contributing to the naturalness of the synthesized waveform.

$$L_{adv}(D_j, G) = \sum_{\hat{\mathbf{x}}} \|1 - D_j(\hat{\mathbf{x}})\|_2^2, \quad (2)$$

$$L_D(D_j, G) = \sum_{\mathbf{x}} [\|1 - D_j(\mathbf{x})\|_2^2 + \|D_j(\hat{\mathbf{x}})\|_2^2] \quad (3)$$

To guide the generator in producing realistic audio, adversarial loss is used. Each sub-discriminator D_j aims to minimize an adversarial loss, $L_{adv}(D_j, G)$, given in Eqn.2 which penalizes the generator when the discriminator can distinguish between real and generated audio. This loss function is calculated as the squared difference between 1 and the discriminator's prediction for the generated audio, encouraging the generator to improve until the discriminator cannot reliably distinguish it from real audio. As the generator minimizes this adversarial loss, it becomes better at producing realistic and indistinguishable audio.

The discriminators are trained with their loss function, $L_D(D_j, G)$, given in Eqn.3 which helps them accurately classify real and generated samples. This loss combines the squared difference between the discriminator's predictions for real and generated samples. By optimizing this loss, the discriminators become more adept at identifying subtle artifacts in generated audio, which in turn pushes the generator to refine its output. This adversarial setup, where the generator and discriminators are trained in opposition, is crucial for producing high-quality audio with minimized artifacts.

In addition to adversarial losses, the generator is guided by a reconstruction loss, which measures the difference between the Mel-spectrograms of the input and generated signals. This reconstruction loss,

$L_{recon}(G)$, given in Eqn.4 is calculated as the $L1$ distance between the Mel-spectrogram of the real and generated audio. By minimizing this difference, the generator learns to capture the spectral characteristics of the original signal, resulting in more accurate timbre and clarity in the synthesized audio. This loss term is essential for ensuring that the generated audio maintains the tonal qualities of the target speaker.

$$L_{recon}(G) = \sum_{\mathbf{x}} \|\phi(\mathbf{x}) - \phi(\hat{\mathbf{x}})\|_1 \quad (4)$$

The feature matching loss, $L_{fm}(D_j, G)$, given by Eqn.5 is another critical component that enhances the quality of the generated audio. This loss measures the similarity between the internal activations of the

$$L_{fm}(D_j, G) = \sum_{\mathbf{x}} \sum_{i=1}^R \frac{1}{M_i} \|\psi_i(\mathbf{x}) - \psi_i(\hat{\mathbf{x}})\|_1 \quad (5)$$

Discriminators for real and generated audio. By matching these activations, the generator aligns its output more closely with the internal representations of real audio as perceived by the discriminators. This loss helps to reduce unnatural artifacts in the synthesized audio and contributes to a more polished and realistic output by making the generated audio exhibit characteristics closer to genuine audio.

The final loss functions for the generator and discriminators combine these various components. For the generator, the multi-objective loss $L_{multi}^G(D, G)$ is given by Eqn. 6,7 is a weighted combination of adversarial, feature matching, and reconstruction losses, with weight parameters $\lambda_{fm} = 2$ and $\lambda_r = 45$. This combination ensures that the generator focuses on producing realistic audio while maintaining the desired spectral characteristics and reducing artifacts. The discriminator's loss $L_{multi}^D(D, G)$ combines individual losses across all sub-discriminators

to improve their ability to discern real and generated audio, enhancing the generator’s performance through adversarial training. Together, these carefully designed loss functions enable the HiFi-GAN vocoder to synthesize high-fidelity, speaker-specific audio from discrete representations.

$$L_G^{multi}(D, G) = \sum_{j=1}^J [L_{adv}(G, D_j) + \lambda_{fm} L_{fm}(G, D_j)] \quad (6)$$

$$+ \lambda_r L_{recon}(G),$$

$$L_D^{multi}(D, G) = \sum_{j=1}^J L_D(G, D_j), \quad (7)$$

3.4 SPEECH RESYNTHESIS BY VOICE CONVERSION

The second methodology builds upon the first by enhancing voice conversion using discrete speech units, optimizing the process for dysarthric speech resynthesis. Unlike the initial approach, which primarily validated the feasibility of HuBERT-derived discrete units for speech reconstruction, this methodology introduces a Speech-to-Unit model that translates dysarthric speech into normal speech representations before waveform synthesis. As shown in Figure 3.5, this pipeline follows a more novel approach that involves parallel training of dysarthric and normal speech units, ensuring that the learned representations are better aligned for intelligibility enhancement.

The process begins with HuBERT-based unit extraction, where normal speech is processed through k-means clustering to generate discrete speech units, which serve as reference targets for the

transformation of dysarthric speech. The dysarthric speech input is then mapped to these reference units through a seq2seq Speech-to-Unit (S2UT) model, which is trained using cross-entropy loss to minimize the divergence between predicted and target speech units. The transformed soft-unit representations are subsequently refined using a modified M-HuBERT model, ensuring that the generated speech maintains both intelligibility and speaker identity. Finally, these estimated soft units are fed into a HiFi-GAN vocoder, which synthesizes the final waveform using a series of dilated convolutional residual blocks, producing high-fidelity speech output.

3.4.1 Encoder

The encoder block is responsible for extracting meaningful representations from both normal and dysarthric speech to facilitate effective speech conversion. The process begins with HuBERT-based unit extraction, where normal speech is passed through a pre-trained HuBERT model that applies k-means clustering to derive discrete speech units. These units capture linguistic content, pitch, and speaker identity in a compact representation, serving as the reference targets for learning transformations from dysarthric speech.

Simultaneously, dysarthric speech is processed in parallel to establish a mapping between its noisy, impaired representations and the corresponding healthy speech units. To ensure smoother transitions and better phonetic alignment, the extracted discrete units are passed through a pre-initialized soft encoder, which converts them into soft continuous units. This step is crucial as discrete representations, while effective, may introduce quantization artifacts that degrade naturalness. The soft encoder mitigates this issue by learning a probabilistic mapping of speech units,

preserving fine articulatory details, and ensuring that the transformed representations remain intelligible and speaker-consistent.

3.4.2 Speech to Unit Model (S2ut Model)

The Speech-to-Unit (S2UT) model plays a critical role in transforming dysarthric speech into intelligible speech representations while preserving speaker identity. Unlike traditional ASR-TTS pipelines that require explicit text transcriptions, this model directly learns a speech-to-speech mapping, making it particularly effective for low-resource languages and dysarthric speech processing. The S2UT model is designed as a sequence-to-sequence (seq2seq) framework, leveraging a Conformer-based architecture that efficiently captures both local phonetic features and long-term speech dependencies. The input to the model consists of Mel spectrogram representations of dysarthric speech, which are processed through convoluted dilated residual blocks (ResBlocks) to expand the receptive field without increasing computational complexity. These blocks help the model learn both short-term articulatory features and long-term speech patterns, making it more effective in handling the irregular prosody and phoneme distortions characteristic of dysarthric speech.

A key feature of the S2UT model is the use of M-HuBERT, a modified self-supervised speech model that refines unit representations by ensuring they align closely with their corresponding healthy speech targets. The model is optimized using a cross-entropy loss function, which measures the difference between the predicted unit distributions and their reference healthy speech units, ensuring accurate phoneme alignment. Additionally, contrastive loss regularization is applied to further enhance the model's robustness across different dysarthria severity levels. This contrastive learning technique ensures that the predicted soft-unit

representations remain distinct from other dysarthric speakers while aligning closely with their corresponding healthy speech representations, thus improving intelligibility.

A parallel training strategy is employed to ensure that the model effectively generalizes across varying dysarthric severities while preserving speaker identity and phonetic integrity. In this setup, normal speech is first processed through a HuBERT-based feature extractor, where k-means clustering is applied to derive discrete unit representations that act as reference targets. Simultaneously, dysarthric speech is fed into the S2UT model, which is trained to predict the corresponding normal speech units. The difference between the predicted and reference units is minimized using cross-entropy loss, while contrastive loss ensures that dysarthric representations remain distinct yet closely aligned with their respective healthy speech counterparts.

To enhance learning, the model integrates convoluted dilated residual blocks, which expand the receptive field and improve the ability to capture long-range phonetic dependencies. This architectural design is particularly useful for modeling speech rhythm, articulation patterns, and phonetic transitions, ensuring that the synthesized speech output is more natural and coherent. Furthermore, the inclusion of M-HuBERT in the training process refines the learned unit representations, preserving articulatory nuances while filtering out unwanted distortions. By leveraging self-supervised learning on multilingual datasets, M-HuBERT enhances the generalization capability of the speech units, making them more robust across different dysarthria severities.

The system setup involves fine-tuning and adapting multiple models for dysarthric speech processing and resynthesis. The model,

containing 95M parameters and 12 transformer layers, was fine-tuned on healthy Tamil speech for 50 epochs using the AdamW optimizer (learning rate = $3e-5$, batch size = 32) to enhance its adaptation for Tamil phonetic structures. To obtain discrete speech units, k-means clustering ($K = 1000$) was applied on the 11th-layer features of mHuBERT, serving as a reference phonetic space for soft-unit encoding. A linear projection layer was then trained for 25 epochs using Adam ($lr = 1e-4$, batch = 64) with a cross-entropy loss, ensuring speaker invariance while preserving phonetic detail.

The Speech-to-Unit Translation (S2UT) model was implemented as a Transformer-based encoder-decoder architecture, consisting of 12 encoder layers and 6 decoder layers. The encoder utilized 4 attention heads, 256-dimensional embeddings, and a Conv1D subsampling module, while the decoder contained 8 attention heads with shared input-output embeddings. The model was trained for 235 epochs ($\sim 36K$ steps) using Adam ($lr = 5e-4$, $\beta_1 = 0.9$, $\beta_2 = 0.98$, 10K warmup steps), optimized with a cross-entropy and contrastive loss ($\lambda = 0.1$) to enhance robustness across dysarthria severities. SpecAugment (frequency masking $F = 10$, time masking $T = 50$) was applied as a regularization strategy, and the full training process took approximately 72 hours.

The parallel training framework significantly improves the ability of the system to learn structured mappings between dysarthric and normal speech, enabling the synthesis of intelligible and natural-sounding speech. The final output of the S2UT model consists of estimated soft units, which are later processed by the HiFi-GAN vocoder to generate high-fidelity speech.

3.4.3 Decoder

The final block in the pipeline represents the inference phase, where the estimated speech units from the Speech-to-Unit (S2UT) model are transformed into natural-sounding speech using the Unit HiFi-GAN vocoder. These estimated units serve as a refined representation of the original dysarthric speech, ensuring improved intelligibility and phonetic accuracy. The Unit HiFi-GAN vocoder is responsible for converting these soft-unit representations into an audio waveform that closely resembles natural speech. The vocoder operates by upsampling the unit representations through convolutional layers and dilated residual blocks, progressively refining the speech output to match the target sampling rate. Additionally, adversarial training with multi-period and multi-scale discriminators ensures that the generated waveform captures both fine-grained phonetic structures and broader speech prosody, leading to more realistic and expressive speech output. The final result is the resynthesized speech, which is significantly clearer and more natural than the original dysarthric input.

For speech resynthesis, the HiFi-GAN vocoder was adapted to accept soft-unit inputs, where the soft-unit probability vectors were projected into a continuous embedding space $E \in \mathbb{R}^{1000 \times 256}$ and processed through transposed convolution layers and residual blocks. The vocoder was trained for 5 days (500K steps, batch size = 16, Adam lr = $2e-4$), optimizing adversarial, feature-matching, and Mel-spectrogram l1 losses. All models were implemented using Fairseq and trained on an NVIDIA GeForce RTX 3050 GPU (24 GB VRAM) to ensure efficient computation and processing. By leveraging the unit-based speech conversion process, this block successfully enhances speech intelligibility

while preserving the speaker's unique vocal identity, making it highly beneficial for assistive communication technologies.

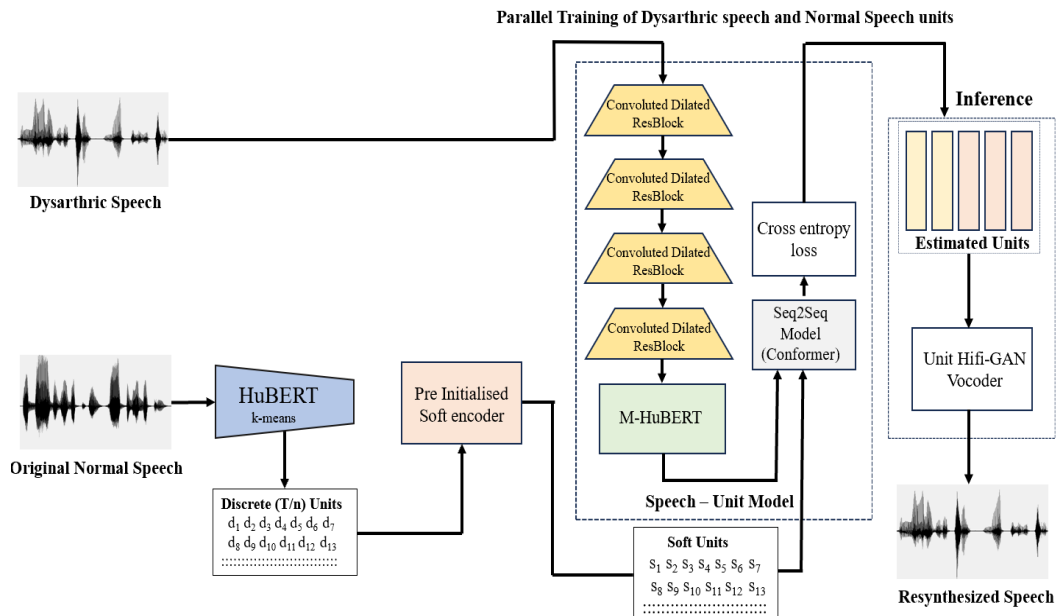


Figure 3.5 End-to-End Dysarthric Voice Conversion Pipeline

3.5 EVALUATING DISENTANGLEMENT VS. VOICE CONVERSION FOR DYSARTHIC SPEECH

The key difference between the two methodologies lies in their approach to speech representation and resynthesis. The first method, based on discrete disentanglement of speech features, primarily focused on validating the feasibility of using HuBERT-derived discrete units for speech reconstruction. It involved extracting discrete content, pitch, and speaker identity representations and using a HiFi-GAN vocoder for synthesis. However, this approach suffered from quantization errors and loss of fine phonetic details, limiting naturalness. The second method, voice conversion based on discrete units, was built upon the first by introducing a Speech-to-Unit Translation (S2UT) model that directly

mapped dysarthric speech units to normal speech units using a seq2seq Conformer-based framework. By leveraging soft-unit modelling and a parallel training strategy, the second method significantly improved speech intelligibility, speaker consistency, and phonetic alignment, making it more effective for real-world applications.

Voice conversion presents a more effective strategy compared to speech disentanglement, particularly for dysarthric speech resynthesis, because it allows for a direct transformation of impaired speech into a more intelligible version while preserving speaker identity. While disentanglement focuses on decomposing and separately manipulating speech attributes, it can introduce errors in unit reassembly and speaker characteristics drift. In contrast, voice conversion directly learns a mapping between dysarthric and normal speech representations, making it more robust to variations in dysarthria severity and requiring less explicit control over individual features. This results in a smoother, more natural speech output without the need for additional phonetic reconstruction steps, making it a more scalable and adaptable approach for speech enhancement in dysarthric speakers.

3.6 EXPERIMENTAL SETUP AND PERFORMANCE ANALYSIS OF THE ENHANCEMENT PIPELINES

3.6.1 Overview

To evaluate the effectiveness of the proposed methodologies, we conducted both subjective and objective analyses to assess speech intelligibility and ASR performance. The subjective evaluation involved Mean Opinion Score (MOS) and Degradation MOS (DMOS) ratings,

where human participants evaluated the intelligibility and perceived quality of the resynthesized speech compared to the original dysarthric speech. This provided insights into the perceptual improvements achieved by both the disentanglement-based method and the voice conversion approach. Additionally, objective evaluations were performed using BERT-based similarity scores, which quantitatively measure the alignment between the resynthesized speech and its corresponding healthy reference speech, enabling a phonetic clarity assessment independent of human perception.

Furthermore, to examine the impact of both methodologies on Automatic Speech Recognition (ASR) performance, we trained three different ASR systems using distinct data configurations. The Baseline ASR system was trained on original dysarthric speech, the Resynthesized ASR system used only the voice-converted speech, and the Combined ASR system incorporated both original and resynthesized speech for training. This setup allowed for comparative analysis, highlighting how speech resynthesis influences ASR performance and whether it improves recognition accuracy for dysarthric speakers. The following section presents a comprehensive evaluation of both methodologies, illustrating their respective strengths and the improvements achieved through voice conversion-based speech resynthesis.

3.6.2 Intelligibility Analysis

In this section, we evaluate the intelligibility of both original dysarthric speech and the resynthesized speech generated through both pipelines. To assess how well the proposed methods improve speech intelligibility, we use two widely recognized evaluation metrics: Mean

Opinion Score (MOS) and Degraded Mean Opinion Score (DMOS). These subjective measures are designed to quantify the perceived quality and intelligibility of speech samples, with MOS reflecting the overall naturalness and clarity, and DMOS focusing on the degradation introduced during the resynthesis process.

3.6.2.1 Mean Opinion Score Evaluation

The Mean Opinion Score (MOS) evaluation as shown in Table 5.1 was conducted to assess the perceived speech quality and intelligibility of the original and resynthesized speech samples. 10 Participants rated the speech samples on a five-point scale, where higher scores indicate better intelligibility and naturalness. The results demonstrate the effectiveness of the voice conversion (VC) pipeline over the disentanglement-based pipeline in both mild and moderate dysarthria cases.

For mild dysarthria, the MOS for the original speech remains constant at 3.8 ± 0.4 across both methodologies. However, after resynthesis, the disentanglement pipeline achieves a slight improvement to 3.9 ± 0.3 , while the VC pipeline outperforms it with a score of 4.2 ± 0.3 , showing that the VC-based approach enhances the naturalness and intelligibility of mild dysarthric speech more effectively.

Table 3.1 Comparison of MOS Scores for Original and Resynthesized Speech

Methodology	Mild		Moderate	
	Original	Resynthesized	Original	Resynthesized
Disentanglement Pipeline	3.8 ± 0.4	3.9 ± 0.3	2.8 ± 0.5	3.2 ± 0.5
VC Pipeline	3.8 ± 0.4	4.2 ± 0.3	2.8 ± 0.5	3.5 ± 0.4

For moderate dysarthria, where intelligibility is inherently lower, the original MOS is 2.8 ± 0.5 for both methods. After resynthesis, the disentanglement method improves the MOS to **3.2 ± 0.5** , whereas the VC pipeline increases it further to **3.5 ± 0.4** . This suggests that the VC model is better at correcting phonetic distortions and improving clarity in more challenging dysarthric speech scenarios. The overall trend indicates that while both methods improve intelligibility, the VC pipeline consistently outperforms the disentanglement approach, particularly for moderate dysarthric speech, where the difference is more pronounced.

3.6.2.2 Degraded Mean Opinion Score Evaluation

The Degradation MOS (DMOS) evaluation as shown in Table 5.2 was conducted to quantify the level of degradation or unnaturalness introduced during resynthesis. Unlike MOS, lower DMOS scores indicate less degradation and better quality.

Table 3.2 Comparison of DMOS Scores for Original and Resynthesized Speech

Methodology	Mild		Moderate	
	Original	Resynthesized	Original	Resynthesized
Disentanglement Pipeline	3.2 ± 0.6	2.6 ± 0.6	3.6 ± 0.2	3.5 ± 0.4
VC Pipeline	3.2 ± 0.6	2.1 ± 0.5	3.6 ± 0.2	3.3 ± 0.5

For mild dysarthria, the original speech DMOS remains at 3.2 ± 0.6 across both methods. After resynthesis, the disentanglement pipeline reduces the DMOS to **2.6 ± 0.6** , while the VC pipeline achieves a better reduction to **2.1 ± 0.5** . This suggests that the VC-based approach introduces fewer artifacts and maintains the integrity of the speech more effectively.

For moderate dysarthria, where speech quality is already degraded, the original speech has a higher DMOS of 3.6 ± 0.2 . After resynthesis, the disentanglement model reduces it to **3.5 ± 0.4** , whereas the VC pipeline further lowers it to **3.3 ± 0.5** . Although the improvement is smaller, the VC model still performs better in reducing degradation, indicating that it better preserves natural speech characteristics even in more severe dysarthric speech. Overall, the VC pipeline consistently yields lower DMOS scores, demonstrating that it produces higher-quality, less degraded speech compared to the disentanglement-based approach.

3.6.3 Subjective Analysis

The BERT-based similarity scores as shown in Table 5.3 were computed to measure the phonetic and linguistic similarity between the resynthesized speech and its corresponding healthy reference speech. A higher score indicates better alignment with natural speech, reflecting improved phonetic accuracy and intelligibility.

Table 4.3 Comparison of BERT Scores for Original and Resynthesized Speech

Methodology	Mild		Moderate	
	Original	Resynthesized	Original	Resynthesized
Disentanglement Pipeline	0.76 ± 0.03	0.81 ± 0.03	$0.55 \pm .04$	0.65 ± 0.05
VC Pipeline	0.76 ± 0.03	0.89 ± 0.02	$0.55 \pm .04$	0.75 ± 0.03

For mild dysarthria, the original speech starts with a similarity score of 0.76 ± 0.03 for both methods. After resynthesis, the disentanglement pipeline achieves a small improvement to **0.81 ± 0.03** , while the VC pipeline significantly outperforms it with a similarity score of **0.89 ± 0.02** . This highlights that the VC approach is far more effective at reconstructing speech that aligns with natural phonetic structures.

For moderate dysarthria, the original similarity score is lower at 0.55 ± 0.04 , reflecting the higher severity of distortions in speech articulation. After resynthesis, the disentanglement model improves the score to **0.65 ± 0.05** , whereas the VC pipeline achieves a much larger improvement to **0.75 ± 0.03** . This further confirms that voice conversion effectively corrects phonetic errors and enhances alignment with natural

speech patterns, making it a superior approach for dysarthric speech reconstruction.

3.6.4 ASR Performance Results

To build the ASR system, we first constructed monogram and trigram language models using Kaldi, a widely used speech recognition toolkit. These language models were then utilized to train a DNN-HMM-based ASR system, which served as the foundation for evaluating the impact of voice conversion on automatic speech recognition performance. The training set consisted of 292 utterances, while 73 utterances were reserved for testing.

To assess the effectiveness of the proposed voice conversion methodology, we conducted three sets of experiments, each comparing the Word Error Rate (WER) across different ASR training configurations:

- (i) **Baseline ASR System:** The ASR model was trained using only the original dysarthric speech data. This setup provides a reference to measure how well an ASR system performs when trained directly on impaired speech without any resynthesis.
- (ii) **Resynthesized ASR System:** The ASR model was trained exclusively on the resynthesized speech generated from the voice conversion pipeline. This experiment evaluates how well ASR performs when trained entirely on enhanced speech representations instead of raw dysarthric speech.
- (iii) **Baseline + Resynthesized ASR System:** The ASR model was trained on a combined dataset consisting of both original dysarthric speech and resynthesized speech. This approach aims

to determine whether combining both forms of speech helps in improving ASR generalization and recognition accuracy.

Table 4.4 Comparison of ASR Performance (WER) for Different Training Setups

Experimental Setup	Mild (WER %)	Moderate (WER%)
Baseline	9.71	19.54
Resynthesized	6.16	13.27
Baseline + Resynthesized	5.73	9.44

Since the subjective and objective analyses demonstrated that the voice conversion (VC) pipeline significantly improved speech intelligibility, the ASR performance was evaluated exclusively using the resynthesized speech generated from the VC approach. The Word Error Rate (WER) was used as the primary evaluation metric, where a lower WER indicates better ASR performance. The results in Table 5.4 show a clear improvement in WER when utilizing resynthesized speech for ASR training.

The Baseline ASR System, trained solely on original dysarthric speech, achieved a WER of **9.71%** for mild dysarthria and **19.54%** for moderate dysarthria. The higher WER for moderate dysarthria highlights the increased difficulty in recognizing speech with more severe articulation impairments. When training the ASR model exclusively on resynthesized speech from the VC pipeline, there was a significant reduction in WER, with **6.16%** for mild dysarthria and **13.27%** for moderate dysarthria, demonstrating that voice-converted speech is more intelligible and better suited for ASR training compared to raw dysarthric speech.

The best results were obtained from the Baseline + Resynthesized ASR System, which was trained on a combined dataset of original dysarthric and resynthesized speech. This setup further improved WER, achieving **5.73%** for mild dysarthria and **9.44%** for moderate dysarthria. The improvement indicates that combining both raw and enhanced speech allows the ASR model to generalize better across different speech impairments, leading to higher recognition accuracy and robustness. These findings validate voice conversion as an effective speech enhancement strategy for improving ASR accessibility for dysarthric speakers.

3.7 SUMMARY

This chapter presented two methodologies for dysarthric speech resynthesis, highlighting their differences and improvements. The first method, based on discrete disentanglement, demonstrated the feasibility of using HuBERT-derived discrete units for speech reconstruction but faced limitations due to quantization errors and loss of fine phonetic details. Building on this, the second method, a voice conversion-based approach, introduced a Speech-to-Unit Translation (S2UT) model that mapped dysarthric speech to normal speech units using a seq2seq Conformer architecture. By incorporating soft-unit modeling, parallel training, and a HiFi-GAN vocoder, this approach significantly improved speech intelligibility and speaker identity preservation. The transition from disentanglement to voice conversion proved to be a more effective and scalable strategy, ensuring clearer, more natural speech resynthesis for dysarthric speakers.

CHAPTER 4

CONCLUSION AND FUTURE ENHANCEMENTS

4.1 CONCLUSION

This study presented an end-to-end dysarthric voice conversion system aimed at enhancing speech intelligibility and improving automatic speech recognition (ASR) performance for individuals with dysarthria. Two methodologies were explored: discrete disentanglement of speech features and voice conversion using discrete units, with the latter proving to be a superior approach. Through subjective and objective analyses, it was observed that voice conversion significantly improved the intelligibility, quality, and phonetic alignment of dysarthric speech, as evidenced by higher MOS, DMOS, and BERT similarity scores compared to the original speech.

Furthermore, ASR evaluations demonstrated that resynthesized speech trained on the VC pipeline led to a substantial reduction in Word Error Rate (WER), confirming the effectiveness of the proposed approach in making dysarthric speech more recognizable for ASR systems. The best ASR performance was achieved using a combined dataset of original and resynthesized speech, demonstrating that data augmentation through voice conversion can improve model generalization and robustness. These results validate voice conversion as a practical and scalable solution for improving speech intelligibility and accessibility for individuals with dysarthria, particularly in low-resource languages where labeled dysarthric speech data is limited.

4.2 FUTURE ENHANCEMENTS

While the proposed voice conversion system has demonstrated significant improvements in speech intelligibility and ASR performance, several areas can be explored for further enhancement. One key direction is speaker adaptation for personalization, where fine-tuning the model to better capture individual speaker characteristics could result in more natural and personalized speech synthesis for dysarthric individuals. Additionally, real-time deployment of the system in augmentative and alternative communication (AAC) devices would make it more accessible for daily use, allowing individuals with dysarthria to communicate more effectively in real-world scenarios.

Further refinements in dysarthric-to-normal speech mapping could also improve speech clarity by integrating contrastive learning or reinforcement learning techniques, enabling the system to learn more robust phonetic transformations. Expanding the model to support multilingual and cross-language adaptation would be another crucial enhancement, allowing voice conversion for dysarthric speakers across diverse low-resource languages. Lastly, an end-to-end ASR integration could be developed by combining voice conversion and ASR training into a unified framework, optimizing both tasks simultaneously for seamless speech recognition improvement. These enhancements would help in scaling the system for broader accessibility, making automatic speech processing more inclusive for individuals with speech impairments.

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